JINSEO KIM

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Seoul, South Korea

PROFILE

Software Engineering undergraduate with research experience in robust object detection, medical image analysis, recommendation systems, and multimodal biosignal classification. Interested in HCI, wearable sensing, multimodal learning, and situated spatial intelligence.

Research Interests: Human-Computer Interaction, Wearable Computing, Multimodal Learning, Spatial Intelligence, Computer Vision

EDUCATION

Chung-Ang University | Seoul, South Korea **Mar 2024 - Feb 2028**

B.S. in Software Engineering (Expected) | GPA: 4.17 / 4.5

- Declared additional majors: Financial AI Convergence and Next-Generation Semiconductor Engineering.

The University of Hong Kong | Hong Kong **Aug 2026 - Jun 2027**

Incoming Exchange Student, School of Computing and Data Science

RESEARCH EXPERIENCE

Undergraduate Research Intern | IXLab, Chung-Ang University **Jul 2026 - Present**

- Optimizing machine-learning and deep-learning pipelines for five-class viewing-distance and nine-class gaze-direction classification from six-channel EOG data collected from 20 participants.
- Implemented and compared classical machine-learning models with 2D CNN and Transformer architectures for raw and feature-based signal classification.
- Evaluated participant-independent generalization using leave-one-subject-out cross-validation and analyzed preprocessing, channel-wise normalization, and temporal segmentation choices.

Undergraduate Research Intern | KELab, Chung-Ang University **Mar 2026 - Jun 2026**

- Conducted research on context-aware travel recommendation by modeling user intent and travel context from online review data.
- Collected and preprocessed Reddit travel-review data in Python and reviewed recent research on recommendation systems and user modeling.

PUBLICATION

E. Lee, **J. Kim**, J. Chung, G. Lim, and J. Paik. "Domain Adaptation-Driven Single Domain Generalization for Thermal UAV-Based Missing Person Detection." **2026 IEIE Summer Conference**, 2026. Presented; second author.

JINSEO KIM

SELECTED RESEARCH PROJECTS

Thermal UAV Missing-Person Detection under Unseen Domains
2026

WBridge Research Program (Government-funded)

- Co-developed a YOLOv5-based single-domain generalization framework using multi-condition Fourier Domain Adaptation, thermal contrast augmentation, and detection-stabilization losses.
- Trained only on HIT-UAV and evaluated without target-domain fine-tuning on the unseen VTSaR and WiSARD search-and-rescue datasets.
- Improved mAP50 from 3.8 to 21.7 on VTSaR and from 12.4 to 23.2 on WiSARD.

Clinically Reliable Synthetic Chest X-Ray Dataset Construction
2026

- Built a multi-agent validation workflow for RoentGen-v2 and Stable Diffusion-generated MIMIC-CXR samples targeting low-prevalence findings.
- Reduced false acceptance of invalid synthetic pairs from 74.4% to 2.1% while retaining 26% high-quality pairs.
- Improved pneumothorax classification AUPRC from 0.18 to 0.27 and recall from 0.41 to 0.58 through validated synthetic augmentation.

TECHNICAL SKILLS

Programming: Python, Java, C/C++, SQL

Machine Learning / Deep Learning: PyTorch, scikit-learn, CNN, Transformer, YOLOv5, XGBoost, Random Forest, MLP

Data & Evaluation: NumPy, pandas, data preprocessing, LOSO cross-validation, ablation studies, mAP, AUPRC, confusion-matrix analysis

Tools: Git, Linux/Ubuntu, CUDA, Jupyter, VS Code, tmux

CERTIFICATIONS & LANGUAGES

Advanced Data Analytics Semi-Professional (ADsP) - Korea Data Agency — **Mar 2026**
SQL Developer (SQLD) - Korea Data Agency — **2026**
English - TOEFL iBT 94
Korean - Native

AWARDS

2nd Place, Signature Competition
Honorable Mention, SUMTECH Hackathon
Excellence Award, Fintech Service Planning Competition — **2025**

LEADERSHIP & ADDITIONAL EXPERIENCE

Department Representative, Software Engineering - Chung-Ang University — **2025**
International Student Mentor - Chung-Ang University — **Mar - Jun 2026**
Coding Instructor - CodingFun Academy — **Jan - Apr 2026**
Team Leader, DB Insurance Financial Competition — **Jan - Feb 2026**